A second method for calculating expected market returns is through a "top-down" approach using an economic growth assumption. Gross domestic product (GDP) is the sum of all goods and services produced or sold in the United States. The Federal Reserve has a target for overall GDP growth, and it attempts to control that growth through changes in monetary policy. That growth number is about 3.0 percent after inflation.

About 10 percent of corporate-generated GDP eventually flows through to corporations as earnings. This number has been fairly consistent over time. Since corporate earnings are ultimately reflected in stock prices, economic growth forecasts can be used to forecast stock returns.

A simple formula for expected stock market returns using the earnings growth method is as follows:

Equity return = earnings per share growth + cash dividends + valuation change

To better understand this model, each variable needs to be explained:

1. *Earnings growth.* The primary driver of long-term stock market gains is corporate earnings. The more money companies earn, the higher the stock market goes relative to inflation. Earnings are a derivative of GDP growth.

2. *Cash dividends.* Many U.S. corporations pay out a portion of their earnings in the form of cash dividends. At the end of 2009, the dividend on the total U.S. stock market was about 2 percent. The growth of dividend payments has been about 3 percent per year, which is slightly less than the growth in earnings. The growth varies because of capitalization issues and decisions made in corporate boardrooms, which extends in part to government policies on taxation. Although higher dividends are attractive to income-seeking investors, there is no free lunch. Higher dividend payout means less cash for corporations to invest, consequently lower expected growth rates of earnings in the future. In addition, we have double taxation of dividends in the United States. Corporate earnings are taxed once at the corporate level and again at the individual level if dividends are paid to shareholders.